

SOLVED PRACTICE WORKBOOK

Weather Elements / India Jet Stream-Western Disturbances - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Weather-element chain?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. C. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. D. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.

Answer: A. Explanation: Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Weather-element chain?

A. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. B. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. C. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. D. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated.

Answer: B. Explanation: Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Weather-element chain?

A. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. B. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. C. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. D. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value.

Answer: C. Explanation: Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Weather-element chain?

A. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place.

Answer: D. Explanation: Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. The other

options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Insolation controls?

A. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: A. Explanation: Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Insolation controls?

A. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. B. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. C. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: B. Explanation: Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Insolation controls?

A. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs.

Answer: C. Explanation: Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Insolation controls?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.

Answer: D. Explanation: Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Heat-budget distinction?

A. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: A. Explanation: The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Heat-budget distinction?

A. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. B. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. C. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. D. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs.

Answer: B. Explanation: The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Heat-budget distinction?

A. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. B. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. C. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. D. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed.

Answer: C. Explanation: The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Heat-budget distinction?

A. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. B. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. C. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. D. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated.

Answer: D. Explanation: The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Lapse rate and inversion?

A. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. D. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs.

Answer: A. Explanation: Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Lapse rate and inversion?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed.

Answer: B. Explanation: Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Lapse rate and inversion?

A. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. B. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. C. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. D. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars.

Answer: C. Explanation: Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Lapse rate and inversion?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. C. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: D. Explanation: Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Pressure-gradient force?

A. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. B. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars.

Answer: A. Explanation: Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Pressure-gradient force?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography.

Answer: B. Explanation: Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Pressure-gradient force?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. C. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. D. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography.

Answer: C. Explanation: Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Pressure-gradient force?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. C. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. D. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value.

Answer: D. Explanation: Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Coriolis force?

A. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. B. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. C. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. D. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography.

Answer: A. Explanation: Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Coriolis force?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. C. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: B. Explanation: Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Coriolis force?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. C. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: C. Explanation: Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Coriolis force?

A. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. B. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. C. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. D. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed.

Answer: D. Explanation: Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Frictional wind?

A. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. B. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. C. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. D. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography.

Answer: A. Explanation: Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Frictional wind?

A. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. B. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. C. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: B. Explanation: Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Frictional wind?

A. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. B. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. C. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. D. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature.

Answer: C. Explanation: Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Frictional wind?

A. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. B. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. C. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. D. Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs.

Answer: D. Explanation: Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Geostrophic balance?

A. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. B. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. C. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: A. Explanation: Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Geostrophic balance?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. C. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: B. Explanation: Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Geostrophic balance?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Convectonal, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: C. Explanation: Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Geostrophic balance?

A. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. B. Convectonal, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. D. Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars.

Answer: D. Explanation: Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Pressure belts?

A. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. B. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. C. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: A. Explanation: Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Pressure belts?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. C. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: B. Explanation: Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Pressure belts?

A. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. B. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: C. Explanation: Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Pressure belts?

A. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. B. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. D. Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography.

Answer: D. Explanation: Equatorial low, subtropical highs, subpolar lows and polar highs are idealised planetary belts that migrate seasonally and are broken by land-sea heating and topography. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Planetary winds?

A. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. B. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. C. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: A. Explanation: Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Planetary winds?

A. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. B. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. C. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. D. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism.

Answer: B. Explanation: Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Planetary winds?

A. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. B. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. D. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it.

Answer: C. Explanation: Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Planetary winds?

A. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. B. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. C. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. D. Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern.

Answer: D. Explanation: Trade winds, westerlies and polar easterlies connect the pressure belts, but monsoons, cyclones, local winds and seasonal migration complicate the ideal pattern. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Humidity measures?

A. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. B. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. C. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. D. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it.

Answer: A. Explanation: Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Humidity measures?

A. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. B. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. C. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. D. Convectional, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism.

Answer: B. Explanation: Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Humidity measures?

A. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. B. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. C. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. D. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.

Answer: C. Explanation: Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Humidity measures?

A. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. B. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature.

Answer: D. Explanation: Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Condensation and clouds?

A. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. B. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. C. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. D. Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism.

Answer: A. Explanation: Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Condensation and clouds?

A. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. B. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. C. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. D. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.

Answer: B. Explanation: Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Condensation and clouds?

A. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. B. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. C. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. D. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall.

Answer: C. Explanation: Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Condensation and clouds?

A. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. B. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Answer: D. Explanation: Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Rainfall mechanisms?

A. Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. B. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it.

Answer: A. Explanation: Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Rainfall mechanisms?

A. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. B. Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. C. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. D. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India.

Answer: B. Explanation: Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Rainfall mechanisms?

A. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. B. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. C. Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. D. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.

Answer: C. Explanation: Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Rainfall mechanisms?

A. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. B. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. C. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. D. Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism.

Answer: D. Explanation: Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Troposphere depth?

A. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. B. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.

Answer: A. Explanation: The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Troposphere depth?

A. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. B. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall.

Answer: B. Explanation: The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Troposphere depth?

A. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. D. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall.

Answer: C. Explanation: The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Troposphere depth?

A. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. D. The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it.

Answer: D. Explanation: The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains Jet-stream definition?

A. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. D. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India.

Answer: A. Explanation: A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of Jet-stream definition?

A. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. B. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. C. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. D. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.

Answer: B. Explanation: A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for Jet-stream definition?

A. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. D. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.

Answer: C. Explanation: A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning Jet-stream definition?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. C. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. D. A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.

Answer: D. Explanation: A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Winter STWJ role?

A. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. B. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. C. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. D. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.

Answer: A. Explanation: In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Winter STWJ role?

A. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. B. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. C. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. D. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.

Answer: B. Explanation: In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Winter STWJ role?

A. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. B. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. C. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. D. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust.

Answer: C. Explanation: In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Winter STWJ role?

A. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. B. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India.

Answer: D. Explanation: In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains Summer easterly-jet boundary?

A. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. D. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.

Answer: A. Explanation: The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of Summer easterly-jet boundary?

A. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. B. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. C. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. D. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.

Answer: B. Explanation: The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for Summer easterly-jet boundary?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. C. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. D. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.

Answer: C. Explanation: The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning Summer easterly-jet boundary?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall.

Answer: D. Explanation: The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Western-disturbance structure?

A. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. B. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. C. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. D. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place.

Answer: A. Explanation: Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Western-disturbance structure?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust.

Answer: B. Explanation: Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Western-disturbance structure?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. C. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. D. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.

Answer: C. Explanation: Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Western-disturbance structure?

A. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. B. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.

Answer: D. Explanation: Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains IMD observation architecture?

A. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. B. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. C. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. D. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place.

Answer: A. Explanation: IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. The other options describe different processes, locations, scales or governance categories.

Q74. Which option is the safest spatial interpretation of IMD observation architecture?

A. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. B. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated.

Answer: B. Explanation: IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. The other options describe different processes, locations, scales or governance categories.

Q75. Which statement preserves the process boundary for IMD observation architecture?

A. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. B. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. C. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. D. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated.

Answer: C. Explanation: IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. The other options describe different processes, locations, scales or governance categories.

Q76. Which option avoids the main UPSC trap concerning IMD observation architecture?

A. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. B. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. C. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. D. IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.

Answer: D. Explanation: IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Verified weather PYQ routes?

A. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. B. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. C. Insolation, temperature, pressure, wind, humidity, clouds and precipitation are linked state variables and processes; weather is their short-period atmospheric expression at a place. D. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.

Answer: A. Explanation: Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Verified weather PYQ routes?

A. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. B. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. C. Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: B. Explanation: Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Verified weather PYQ routes?

A. The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated. B. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. C. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. D. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.

Answer: C. Explanation: Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Verified weather PYQ routes?

A. Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value. B. Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants. C. Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed. D. Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust.

Answer: D. Explanation: Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Verified routing retains direct owner demands on weather elements and upper-air processes, including the 2021 and 2022 GS-I questions and relevant objective routes. No official answer letter is invented where the ledger withholds it, and no unrelated Disaster Management warning question is relabelled direct.

OWNER PYQ LEDGER EXTRACTS

□ Temperature Inversion (PYQ / high-yield gap)

FACT: Grounded in GC Leong + Majid Husain + ANALYSIS: standard geography. Added to close a UPSC gap.

- FACT: **Temperature inversion** is reversal of the normal lapse rate: air near the surface becomes colder than air above, as shown in book passages on valley-bottom inversion and low-level inversion with frost.
- FACT: Favourable conditions: long winter nights, cloudless clear sky, dry air/low relative humidity and calm or slow air movement.
- ANALYSIS: Mechanism: nocturnal radiational cooling chills the ground; dense cold air drains into valleys/frost hollows and gets trapped below warmer air.
- ANALYSIS: India link: winter inversions over the Indo-Gangetic plains can trap fog, smoke and pollutants near the surface.

Type	Mechanism	Exam association
FACT: Radiation inversion	Night-time surface cooling	Frost, fog, pollution trapping
ANALYSIS: Valley inversion	Cold-air drainage downslope	Frost hollows, colder valley floor
ANALYSIS: Subsidence inversion	Descending anticyclonic air warms aloft	Stable dry layer, smog cap

MEMORY: Trap: Inversion is **stability**, not instability; it suppresses vertical mixing even if pollution/fog near the ground looks dramatic.

□ Katabatic & Anabatic Winds (local-winds PYQ gap)

FACT: Grounded in Majid Husain + ANALYSIS: standard geography. Added to close a UPSC gap.

- FACT: Mountain-valley winds reverse direction daily: book passages state daytime heating makes air rise up mountain slopes as a warm **valley wind**.
- ANALYSIS: **Anabatic wind** = daytime upslope flow caused by slope heating; **katabatic wind** = night-time downslope drainage of cold, dense air.
- ANALYSIS: These are local, thermally driven winds, like sea breeze/land breeze logic, but controlled by relief instead of shoreline contrast.
- ANALYSIS: India link: Himalayan valleys show daytime upslope cloud build-up and night-time cold-air pooling that aids frost/fog.

Wind	Time	Direction	Thermal character
ANALYSIS: Anabatic / valley wind	Day	Upslope	Warm rising air
ANALYSIS: Katabatic / mountain wind	Night	Downslope	Cold dense drainage
FACT: Sea breeze	Day	Sea -> land	Cooler onshore air
FACT: Land breeze	Night	Land -> sea	Offshore drainage

MEMORY: Trap: A **valley wind** is named from its origin but blows **upslope** in daytime; a **mountain wind** blows downslope at night.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2026.md. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	Prelims GS-I	84	Bharat Forecast System's resolution objective and developing institution	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (Ans-2026-GS1-Provisional); key is provisional - no answer letter recorded or inferred here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Bharat Forecast System's resolution objective and developing institution

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2024-2025.md, _PYQ-ROUTING-PRELIMS-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 11

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	GS-I	4	Sea surface temperature rise and formation of tropical cyclones	Explain · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	GS-I	6	The phenomenon of cloudbursts	Explain · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	GS-I	16	Twisters and their concentration around the Gulf of Mexico	Explain · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	Prelims GS-I	1	Atmosphere heated more by solar than terrestrial radiation; greenhouse gases	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	2	Thickness of the troposphere at the equator versus poles; convection	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	4	Inferences from January isothermal maps	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	12	Water-vapour characteristics with altitude	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	14	Coriolis force characteristics	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	26	Atmospheric dust distribution across climatic zones	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	27	January isotherms bending over land and ocean	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	29	Atmosphere maintaining Earth's temperature; CO ₂ absorbing radiation	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Sea surface temperature rise and formation of tropical cyclones
- The phenomenon of cloudbursts
- Twisters and their concentration around the Gulf of Mexico
- Atmosphere heated more by solar than terrestrial radiation; greenhouse gases
- Thickness of the troposphere at the equator versus poles; convection
- Inferences from January isothermal maps
- Water-vapour characteristics with altitude
- Coriolis force characteristics
- Atmospheric dust distribution across climatic zones
- January isotherms bending over land and ocean
- Atmosphere maintaining Earth's temperature; CO₂ absorbing radiation

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md, _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2020, 2021, 2022, 2023

- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands: 7**

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	44	Absence of dewdrops formation on cloudy nights	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	Prelims GS-I	99	Jet streams hemisphere limitation and cyclone eye temperature	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	GS-I	14	Alignment of major mountain ranges and impact on local weather	Briefly mention and explain · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-I	17	Troposphere as the layer determining weather processes	How · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	Prelims GS-I	81	High and low clouds differential effects on Earth temperature	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	62	Insolation distribution in Earth's atmosphere infrared rays	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	64	Temperature contrast between continents and oceans seasons	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Absence of dewdrops formation on cloudy nights
- Jet streams hemisphere limitation and cyclone eye temperature
- Alignment of major mountain ranges and impact on local weather
- Troposphere as the layer determining weather processes
- High and low clouds differential effects on Earth temperature
- Insolation distribution in Earth's atmosphere infrared rays
- Temperature contrast between continents and oceans seasons

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

□ Rossby Waves (jet-stream / Western Disturbance PYQ gap)

FACT: Grounded in Majid Husain + D.R. Khullar + ANALYSIS: standard geography. Added to close a UPSC gap.

- FACT: Majid Husain links Rossby-wave/geostrophic upper-air flow with winds moving parallel to isobars where pressure-gradient force balances Coriolis force in the upper troposphere.
- FACT: D.R. Khullar's jet-stream passages note that the upper jet steers western depressions from the Mediterranean towards India.
- ANALYSIS: **Rossby waves** are large meanders in the westerly upper-air flow caused by Earth's rotation and latitudinal variation in Coriolis force.
- ANALYSIS: India link: stronger meanders/blocking can alter Western Disturbance tracks, cold waves, heat domes and pre-monsoon rainfall.

Feature	Straight/zonal jet	Wavy/meridional Rossby flow
ANALYSIS: Path	West-east, faster	Large north-south loops
ANALYSIS: Weather	Quick-moving systems	Blocking, persistent extremes
FACT: India link	WD conveyor	WD track shifts / stalled systems

MEMORY: Trap: Rossby waves are **upper-air planetary waves**, not ocean waves; they steer weather systems rather than directly producing rain by themselves.

PYQ DEMAND CARD 1 - 2021 GS-I

Demand: Briefly mention the alignment of major mountain ranges of the world and explain their impact on local weather conditions, with examples. (250 words)

Status: Verified routed direct Mains demand.

Model solution: Map east-west and north-south ranges, then explain barrier, orographic rain, rain shadow, channelling, foehn-type descent and cold-air blocking. Use Himalaya and Western Ghats examples and qualify by synoptic wind direction, pass geometry and season.

PYQ DEMAND CARD 2 - 2022 GS-I

Demand: Discuss how the troposphere determines weather processes. (250 words)

Status: Verified routed direct Mains demand.

Model solution: Link surface heating, water vapour, lapse rate, convection, condensation, pressure systems and upper-air steering within the troposphere. Add its greater tropical depth and the tropopause boundary. Qualify that stratospheric and oceanic influences can modulate tropospheric weather.

PYQ DEMAND CARD 3 - 2024 Prelims GS-I

Demand: Inferences from January isothermal maps.

Status: Verified routed objective demand; official Set-A key exists locally but the routing ledger records no answer letter.

Model solution: Trace land-sea thermal contrast, current influence and continentality from the mapped isotherms. Preserve the official option order and do not manufacture an answer letter.

PYQ DEMAND CARD 4 - 2025 Prelims GS-I

Demand: Atmospheric dust distribution across climatic zones.

Status: Verified routed objective demand; official Set-A key exists locally but the routing ledger records no answer letter.

Model solution: Use source, aridity, vegetation cover, uplift and washout rather than assuming uniform vertical or latitudinal distribution. Preserve the official option order and do not invent a key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why relative humidity can rise at night without additional water vapour. Answer in about 150 words.

Model thesis: Cooling lowers saturation capacity, so relative humidity can rise toward saturation even when actual vapour content changes little.

Claim -> named evidence -> analysis -> qualification:

- Absolute, specific and relative humidity measure different properties; relative humidity changes with both water-vapour content and temperature.
- Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.

Qualified conclusion: Cooling lowers saturation capacity, so relative humidity can rise toward saturation even when actual vapour content changes little.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate pressure-gradient, Coriolis and frictional controls on wind. Answer in about 150 words.

Model thesis: Pressure gradient initiates acceleration, Coriolis deflects motion, and friction slows and turns surface flow across isobars.

Claim -> named evidence -> analysis -> qualification:

- Horizontal pressure differences accelerate air from higher toward lower pressure, with strength related to gradient rather than the absolute pressure value.
- Earth's rotation deflects moving air to the right in the Northern Hemisphere and left in the Southern Hemisphere, is zero at the equator and increases poleward for a given speed.
- Near the surface, friction slows wind, weakens Coriolis deflection and allows flow to cross isobars toward low pressure; aloft the balance differs.
- Away from the equator and friction layer, pressure-gradient and Coriolis forces can approximately balance, producing wind nearly parallel to straight isobars.

Qualified conclusion: Pressure gradient initiates acceleration, Coriolis deflects motion, and friction slows and turns surface flow across isobars.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain how atmospheric stability and uplift mechanism determine clouds and rainfall. Answer in about 250 words.

Model thesis: Lapse structure controls parcel buoyancy, while convection, terrain and fronts provide lift; moisture and microphysics determine precipitation efficiency.

Claim -> named evidence -> analysis -> qualification:

- Temperature generally decreases upward in the troposphere, while an inversion has temperature increasing with height through a layer and can suppress mixing and trap pollutants.
- Air reaching saturation through cooling or moisture addition can condense on nuclei, forming cloud or fog when lift, stability and droplet processes permit.
- Convective, orographic and cyclonic or frontal rainfall describe uplift mechanisms; one storm can combine more than one mechanism.

Qualified conclusion: Lapse structure controls parcel buoyancy, while convection, terrain and fronts provide lift; moisture and microphysics determine precipitation efficiency.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse why the troposphere is the principal weather-producing layer. Answer in about 250 words.

Model thesis: Its water vapour, vertical mixing, heating from below and pressure-temperature gradients host clouds, storms and upper-air steering, with depth varying by latitude.

Claim -> named evidence -> analysis -> qualification:

- Received solar energy varies with solar angle, day length, Earth-Sun distance, cloud, dust and surface orientation; latitude alone does not determine daily receipt.
- The atmosphere gains energy from absorbed solar radiation and from terrestrial longwave, sensible and latent heat; incoming shortwave and outgoing longwave must not be conflated.
- The troposphere is generally deeper over the warm tropics and shallower toward the poles, and most weather occurs there because water vapour, vertical mixing and instability are concentrated in it.
- A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.

Qualified conclusion: Its water vapour, vertical mixing, heating from below and pressure-temperature gradients host clouds, storms and upper-air steering, with depth varying by latitude.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine the role of upper-air jets and western disturbances in India's seasonal weather. Answer in about 300 words.

Model thesis: Winter STWJ steering and evolving western disturbances support rain and snow, while summer easterly flow belongs to a wider monsoon system; no jet acts as a deterministic switch.

Claim -> named evidence -> analysis -> qualification:

- A jet stream is a narrow current of strong upper-tropospheric or lower-stratospheric winds associated with sharp horizontal temperature gradients; it is not fixed at one altitude.
- In winter the subtropical westerly jet lies south of the Himalaya in the broad Indian-sector circulation and helps steer western disturbances toward north and northwest India.
- The tropical easterly jet is a summer upper-air feature linked to the monsoon circulation, but its appearance or strength is not a single-cause switch that proves monsoon onset or rainfall.
- Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.

Qualified conclusion: Winter STWJ steering and evolving western disturbances support rain and snow, while summer easterly flow belongs to a wider monsoon system; no jet acts as a deterministic switch.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an evidence chain for forecasting western-disturbance impacts over India. Answer in about 300 words.

Model thesis: Combine upper-air soundings, satellite, radar, numerical ensembles, terrain and impact exposure, while separating observation, forecast, warning and verified outcome.

Claim -> named evidence -> analysis -> qualification:

- Western disturbances are eastward-moving mid-latitude troughs, lows or cyclonic circulations embedded in the westerlies and commonly draw moisture along their path; they are not southwest-monsoon systems.
- IMD's official upper-air service describes radiosonde and pilot-balloon observations that support analysis of winds, pressure and temperature aloft; observation, diagnosis and forecast remain distinct.
- Direct routes include 2021 GS-I mountain alignment and local weather, 2022 GS-I tropospheric weather processes, and objective demands on jet streams, clouds, insolation, isotherms and atmospheric dust.

Qualified conclusion: Combine upper-air soundings, satellite, radar, numerical ensembles, terrain and impact exposure, while separating observation, forecast, warning and verified outcome.